BCSE0133: MACHINE LEARNING LAB

Credits: 01 **L–T–P: 0–0–2**

S.No	PROGRAM
1	Estimate parameters of a model based on Linear Regression method using a given set of training data set.
2	Estimate parameters of a model based on maximum likelihood estimation method using a given set of training data set.
3	Compute weights of ANN based on back propagation method using a given training dataset.
4	Compute probability of a person to be diabetic based on a given dataset of diabetic persons using Naïve Bayesian classifier.
5	Classify a person as male or female based on a given dataset using naïve Bayesian Classifier, and calculate accuracy, precision, and recall for your data set.
6	Write a program to implement *k*-Nearest Neighbour method to classify the iris data set. Print both correct and wrong predictions. Use Java/Python ML library classes
7	Predicts whether the bank should approve the loan of an applicant, based on his profile using Ensemble learning method.
8	Apply Ensemble learning to cluster a set of data stored in a .CSV file. Use the same dataset for clustering using *k*-Means method. Compare the results of these two algorithms and comment on the quality of clustering. You can add Java/Python ML library classes/API in the program.
9	The stock prediction data is used to predict, whether the stock will go up or down. Perform the task of feature selection with the help of wrapper method.
10	Identify principal components of Big Mart sales data using Principal component analysis (PCA). Also plot the result of PCA, and give inferences.

Outcome: After studying the subject, the students will be able to:

- Able to use different digital image processing algorithms.
- Design code and test digital image processing applications using MATLAB/Python language.
- Apply image processing algorithms in real-life problems.